

Lujain Al-Momani

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[linkedin.com/in/lujain-almomani](https://www.linkedin.com/in/lujain-almomani) | [Laj-22.github.io/lujainMomani-portfolio/](https://github.com/Laj-22) | Legal Status:Golden Visa Holder

Education

University of Wollongong in Dubai, Bachelor of Engineering in Telecommunication & IoT Dubai, UAE

Experience

V2X Networking Architecture Engineer – Bosch Future Mobility Challenge, Sep 2025 – Present
Dubai, UAE

- Led V2X networking architecture design for autonomous driving algorithms, enabling proactive real-time data exchange with traffic monitors and infrastructure to drive enhanced decision-making, accuracy, and on-track performance.

Research Intern, University of Wollongong (UOWD) – Dubai, UAE Sept 2024 – Present

- Developed an ML-based real time Intrusion Detection System (IDS) for VANETs with 97% accuracy,in resource-constrained environments,published the work.

Business Intelligence Intern, Jordan Payments & Clearance Company (JOPACC) – July 2023 – Aug 2023
Amman, Jordan

- Drove executive-level reporting by designing ETL workflows and leveraging Power BI to deliver real-time, data-driven insights that directly support strategic business decisions.

Publications

- Almomani, L., et al. **A Hybrid FeedForward Neural Network-Genetic Algorithm Approach for Beam Selection Optimization in 5G Massive MIMO Systems**. ICCE 2025.
- Almomani, L., et al. **Beyond Manual Labelling: Enhancing Antenna Classification and Performance with Fuzzy Clustering and Machine Learning**. FICTA 2025.
- Abbasi, M., Almomani, L., Alkhatib, O. **Intrusion Detection in VANETs: A Comparative Study of Machine Learning and Deep Learning Models**. ISDIA 2025.
- Almomani, L., et al. **A Comprehensive Review of Security Mechanisms in VANETs and IoV**. (Under Review)
- Abbasi, M., Almomani, L., Alkhatib, O. **Real-Time Intrusion Detection in VANETs Using Lightweight Machine Learning and MQTT**. (Under Review)

Projects

Stateful Fog Security Gateway for IoT — Engineered an AI-driven dual-layer IPS with custom Snort MQTT parser, rules, and self-curated dataset, delivering 98% real-time accuracy for inline MQTT command injection prevention.

Batterly: Smart Pancake Printer — Designed a viscosity-adaptive pancake printer with AI calibration via OpenCV-based vision system on Raspberry Pi, integrating embedded controls for consistent extrusion and real-time G-code generation.

Autonomous Waste Collection Robot (ERC 2025) — Engineered an autonomous UGV by integrating ROS navigation, a custom forklift mechanism, and a fine-tuned MobileNetV2 model for real-time object detection.

Hardware-Accelerated Intrusion Detection System for VANETs — Prototyped a real-time, hardware-based IDS on a Raspberry Pi 5, achieving 99.72% detection accuracy and validating optimal communication protocols.

Skills & Certifications

Networking, Cybersecurity & Management: Network Architecture & Security, Intrusion Detection Systems (IDS), Fault-Tolerant System Design, Network Protocols, AgenticAI, Agile Project Management, Process Optimization, Technical Research, Strategic Planning, Risk Assessment, Team Leadership

Languages & Technologies: Python, Java, C, MATLAB, ROS, Git, VMware, Kali Linux, Cisco IOS, Machine Learning

Certifications: PWC Elevate Me (2025), McKinsey Forward Program (2024), Agile Software Development (Nano Degree, 2023)

Languages: Arabic, English || **Affiliations:** IEEE, Engineers Australia || **Winning Competitions:** HackmyRobot(NYUAD25),ERC(AUS25),Nexus2.0 (Amity25),YouthxGE23